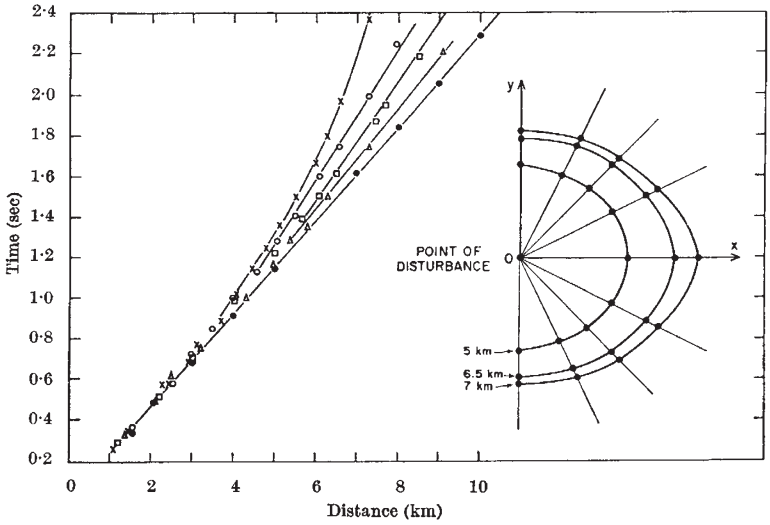


Fig. 2. Travel-time curves and distortion of these from circles to ellipses in an active block. ×, 0° x direction; ○, 30° from x axis; □, 45° from x axis; △, 60° from x axis; ●, 90° from x axis (y direction)

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Palæomagnetic Results from Northern Canada suggesting a Tropical Proterozoic Climate

The Coppermine and September Mountains, in the north-central District of Mackenzie, Northwest Territories of Canada, are composed of a thick succession of tholeiite basalt flows. Directions of natural remanent magnetization were measured on 51 oriented samples from 24 sites (Fig. 1). The sites represent 24 of the more than 60 separate flows occurring in a vertical thickness of more than 5,000 ft. The flows dip from 0° to 5° north in the area sampled. Two potassium/argon whole rock age determinations on flow rocks, made in the Isotope and Nuclear Geology Section of the Geological Survey of Canada (Wanless, private communication), gave an average age of 1,150 m.y. with a probable error of ± 100 (calculated as two standard deviations from the mean). This places the age of extrusion of the flows in upper middle Proterozoic time. The lavas conformably overlie dolomite of the Hornby Bay Group, and in places thin sandstone bands are intercalated between the flows. A sample from a red sandstone intercalation gives a direction of $D = 236$, $I = +24$, which is similar to that of the flows.

Separate specimens from each sample were subjected to magnetic cleaning[1] and thermal cleaning[2] to reduce possible errors from secondary components. The precision parameter[3] of the natural remanent magnetization grouping ($k = 7$, $\alpha = 12$) (Fig. 2*a*) was much improved after treatment in an alternating magnetic field of 75 œrsteds peak value ($k = 23$, $\alpha = 6$) (Fig. 2*c*). The direction after magnetic cleaning was 7° farther from the dipole field direction than that of the natural remanent magnetization. The scatter increased slightly ($k = 5$, $\alpha = 15$) in specimens heated to 480° C, and the resultant moved 13° away from the present field direction. This increase in scatter (Fig. 2*b*) may be due to the heating of about half the specimens above the Curie temperature of their dominant magnetic mineral. The differences between the mean directions of magnetization before treatment, after magnetic cleaning, and after thermal cleaning are not significant at $P = 0{\cdot}05$; but the fact that the cleaned directions are farther from the dipole field direction than before treatment suggests that a small viscous component in the direction of the dipole field has been removed.

Four specimens were heated to 625° C and cooled in a magnetic field of known direction and of intensity 1·3 œrsteds. The resulting thermoremanent magnetization direction was in each case within 2° of the direction of the field during cooling, showing that magnetic anisotropy does not affect the direction of thermoremanent magnetization in the specimens.

The laboratory tests support the idea that the natural remanent magnetization direction is that of the Earth's magnetic field at the time the lavas cooled. Furthermore, this direction is very close to that obtained in a preliminary investigation of gabbroic rocks of the nearby Muskox Intrusion (see Table 1), which has been assigned a radiogenic age of 1,150 m.y.[4] with a probable error of ±40 (calculated as two standard deviations from the mean).

The virtual geomagnetic pole position calculated from the mean direction of the flows (1° N., 176° W.), assuming that the dipole field hypothesis[5] is relevant to middle Proterozoic time, places the flows at a palæolatitude of 10°–15°, and suggests that the Hornby Bay Group with which they are associated was deposited under tropical

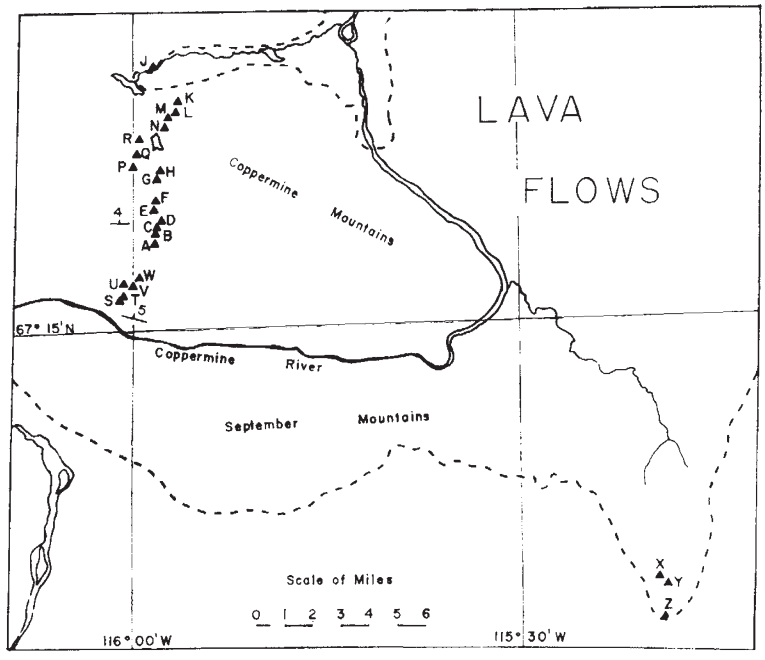


Fig. 1. A sketch map showing sampling sites

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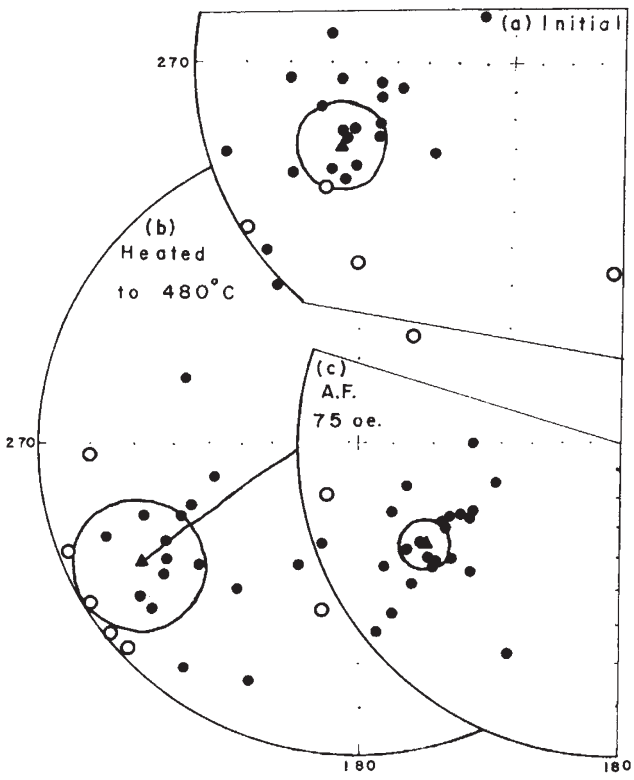


Fig. 2. Portions of stereographic nets showing mean site magnetization directions, (*a*) before treatment, (*b*) after heating to 480° C in a null magnetic field, (*c*) after treatment in an alternating magnetic field of 75 œrsteds peak value. North-seeking directions are plotted as open circles on the upper and dots on the lower hemisphere. Mean group directions, with the trace of the 95 per cent cone of confidence, are shown as triangles. The great circle between the heated mean and the dipole field direction is also shown

conditions. The occurrence of gypsum and anhydrite in the Minto Inlet Formation of Victoria Island, of late Pre-Cambrian age[6], is consistent with such a low latitude.

The close agreement of this pole position with those of comparable age from the Sudbury diabase dykes[7], the Duluth Gabbro[8], and the Muskox gabbroic rocks (Table 2), and the small polar errors ($d\chi = 7$, $d\psi = 4$) associated with the Coppermine lavas pole position, suggest that this virtual geomagnetic pole position may be used as an anchor for the North American polar wandering curve[8,9] for upper middle Proterozoic time.

Table 1. Coppermine Lava Flows and Muskox Gabbro Mean Directions and Statistics

Formation	Treatment	No. sites	No. samples	No. specimens	*D*	*I*	*R*	*k*	*a*
Coppermine lava flows	None	24	51	102	244	+29	20·88	7·4	12
,,	Alternating field of 75 œrsteds	24	48	48	242	+22	23·00	23	6
,,	Heated to 480° C	24	46	46	240	+15	19·11	4·7	15
Muskox gabbroic rocks	None	8	19	38	247	+33	7·48	13	16
,,	Alternating field of 75 œrsteds	8	19	19	242	+29	7·35	11	18

D, Declination east of true north; *I*, inclination positive downwards; *R*, resultant vector using sites as unit vectors; *k*, estimate of precision (ref. 3); *a*, half-angle of the cone of confidence at $P = 0{\cdot}05$ (ref. 3).

Table 2. Canadian Upper Middle Proterozoic Virtual Geomagnetic Pole Positions

Formation	Radiogenic age (m.y.)	Pole position Latitude	Pole position Longitude	Polar errors $d\psi$	Polar errors $d\chi$
Coppermine lavas	1,150 ± 100	1 N.	176 W.	[d] 4	7
				3	6
Muskox gabbroic rocks	[a] 1,150 ± 40 (ref. 4)	4 N.	175 W.	[d] 11	19
				7	12
Duluth Gabbro	[b] 1,120 (ref. 10)	24 N.	180 W.	6	10
Sudbury diabase dykes	[c] 1,200 (ref. 11)	8 S.	162 W.	6	14

a, Average of 4 potassium/argon determinations, 2 from biotite and 2 from whole rock; *b*, average of 2 potassium/argon and 2 rubidium/argon determinations, which range from 1,075 to 1,200; *c*, average of 4 potassium/argon determinations, 2 from biotite and 2 from whole rock; *d*, the larger polar errors for these groups were computed from site means; the remaining polar errors were computed from sample means.

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PHYSICS

Meta-stable Ions in Double-focusing Mass Spectrometers

'Meta-stable peaks' are normally observed due to ions which dissociate near the entrance slit of the magnetic analyser in the sector mass spectrometer. In addition 'meta-stable peaks' due to ions which dissociate within the electrostatic analyser of a double-focusing mass spectrometer can also be observed and their position can be adjusted by varying the voltage across the electrostatic analyser.

Ions which dissociate during their transit of a mass spectrometer analyser tube are said to be meta-stable[1]. They are noted on the mass scale of the spectrum at a position corresponding to m^*, where:

$$m^* = \frac{m_2^2}{m_1}\left\{1 + \frac{(m_1 - m_2)(V - V_1)}{m_2 V}\right\}$$

and m_1 is the mass of the ion before dissociation, m_2 is the mass of the charged fragment after dissociation, V is the accelerating voltage and V_1 is the potential difference through which the ion of mass m_1 falls before decomposing[2]. In general, dissociation can occur anywhere between the ion source and the collector. There may also be a release of internal energy during the transition, so that conditions may be somewhat more complex than indicated by the above equation and many of the ions will not be recorded. A very small internal energy release will discriminate strongly against ions formed at large values of $(V - V_1)$ and prevent their passing through the entrance slit of the spectrometer at the end of the accelerating region. Owing to the directional focusing properties of the sector magnetic field, the decomposing ions most likely to be recorded, in a single-focusing mass spectrometer of the Nier design[3], are those that undergo decomposition near the entrance slit. In this case :

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